



Assessment of multidrug-resistant profile, multi-locus sequence typing and efflux pump activity in *Salmonella* Typhimurium isolated from hospital sewage

Manoj Kumawat¹ · Bilkees Nabi² · Poonam Sharma¹ · Namrata Pal¹ · Devojit Kumar Sarma¹ · Swasti Shubham¹ · Rajnarayan R Tiwari¹ · Samradhi singh¹ · Manoj Kumar¹

Received: 6 February 2023 / Accepted: 5 April 2023 / Published online: 17 April 2023
© The Author(s), under exclusive licence to Springer Nature B.V. 2023

Abstract

Salmonella enterica serovar Typhimurium is becoming a leading cause of gastroenteritis and mortality. The use of antibiotics has increased natural resistance of *S. Typhimurium* to antibiotics. This study aims to isolate and characterize multi-drug-resistant (MDR) *Salmonella* strains from hospital sewage samples in Bhopal City, central India. The MDR isolates were characterized by molecular identification, antimicrobial resistance patterns, multi-locus sequence typing, and efflux pump activity. Specific genes (*hlyA*, *stn*, *invA*, *typh*, and *iroB*) were used to confirm *S. Typhimurium* isolates. The Kirby-Bauer method was employed to profile antimicrobial resistance using 20 antibiotics. Multi-locus sequence typing confirmed *S. Typhimurium* using seven housekeeping genes (*aroC*, *dnaN*, *hemD*, *hisD*, *purE*, *sucA*, and *thr*). Out of five strains, only four were confirmed as *S. Typhimurium* during MLST analysis. Efflux pump activity was determined using the ethidium bromide (EtBr) cartwheel test. Of the 160 isolates, 38 were presumptively confirmed as *S. Typhimurium* based on biochemical characterization, and only five MDR *Salmonella* strains were selected for their resistance against most antibiotics. Efflux pump activity revealed that five out of the four MDR isolates did not retain EtBr inside the cells, indicating pronounced efflux activity. Additionally, the isolated strains showed a specific correlation between the antimicrobial phenotypes and genotypes. The results of this study provide a better understanding of the characterization of *S. Typhimurium* serotype in Bhopal City. Future studies should focus on understanding changing antimicrobial resistance patterns, pathogenicity, and the genetic background of *Salmonella* serotypes. Further surveillance activities for antimicrobial-resistant *Salmonella* in different environmental sources should be prioritized.

Keywords Multi-drug resistance · Antimicrobial resistance · *Salmonella* Typhimurium · MLST · Efflux pump activity

Introduction

Salmonella is a common foodborne disease worldwide, caused by around 2,500 serovars of *S. enterica* that can harm animals and humans (Mezal et al. 2014; Shah et al. 2017). *Salmonella* Typhimurium (ST) is commonly found

in humans and animals, including livestock, rodents, and birds. Non-typhoidal *Salmonella* (NTS) gastroenteritis is a growing global health concern, with an estimated 94 million cases and 155,000 deaths each year (Majowicz et al. 2010). It is the most common cause of food poisoning among NTS *Salmonella* in most countries, particularly in the Indian subcontinent (Greig and Ravel 2009; Bakshi et al. 2003). Salmonellosis is primarily spread through animal-derived foods, particularly poultry, and poultry products have been demonstrated to serve as a primary vehicle for human salmonellosis (Kumawat et al. 2021). Pigs and poultry are key reservoirs of resistant *Salmonella*, which can be transferred to people through various routes (Wang et al. 2022) (e.g., direct contact, food consumption, the spread of manure).

✉ Manoj Kumar
manoj15micro@yahoo.co.in

¹ Department of Microbiology, ICMR- National Institute for Research in Environmental Health, Bhopal, Madhya Pradesh, India

² Department of Biochemistry & Biochemical Engineering, SHUATS, Allahabad, Uttar Pradesh, India

The injudicious use of antibiotics in the health care system and veterinary results in the emergence of antibiotic resistance and serves as a continuous source of AMR in the environment. Due to the risk of infection caused by antibiotic-resistant pathogens, the emergence of antibiotic-resistant pathogens has become a serious threat to public health (Kumar et al. 2021). The frequent and unrestricted use of antimicrobial drugs in veterinary medicine, especially in food animal production, is one of the major causes of the development and spread of AMR (Marshall and Levy 2011). Among the *Salmonella* strains, the most prevalent MDR serovar recognized so far is *Salmonella enterica* serovar Typhimurium (*S. Typhimurium*) associated with food-producing animals, and it accounted for nearly half of all MDR *Salmonella* isolates in the United States over past 12 years. The emergence of multidrug-resistant pathogenic bacteria is considered as an emerging global health threat by organizations like the Centers for Disease Control and Prevention (CDC), the European Centre for Disease Prevention and Control (ECDC), and the World Health Organization (WHO) (Vidic et al. 2022). A high degree of resistance to ampicillin (69.9%), chloramphenicol (56.3%), streptomycin (72.1%), and tetracycline (87.6%) was reported in *S. Typhimurium* isolated from pigs (Brunelle et al. 2016). According to several investigations, multidrug-resistant (MDR) *Salmonella* has been linked to an increased risk of hospitalization, invasive disease, and mortality (Martin et al. 2004; Varma et al. 2005). MDR is usually connected with the plasmids that carry multiple resistance determinants in transposons and integrons (Fluit 2005; Fluit and Schmitz 1999; Leverstein-van Hall et al. 2003; Wiesner et al. 2009). Multidrug resistance (MDR) in *Salmonella* species can be attributed to a variety of mechanisms, including i. modification of the target site (e.g., via mutations in DNA gyrase and topoisomerase IV, as shown in fluoroquinolone resistance); ii degradation of the antibiotic by hydrolysis or modification of the drug to an inactive form (e.g., lactamases and aminoglycoside resistance, respectively); and iii decreases the effective intracellular concentration of antimicrobial agents, which prevents drugs from accessing their targets (Bharadwaj et al. 2022; Chaudhari et al. 2022).

As a result of the high consumption of antibiotics, the critical illness of patients, and the permanent influx of pathogenic species in healthcare environments, resistance can quickly develop, which creates the perfect environment for the dissemination and horizontal transfer of resistance genes. Compared with traditional methods, molecular typing is highly sensitive, specific, fast, and shows better standardization and reproducibility (Herrera-Leon et al. 2007). The multi-locus sequence typing (MLST) method uses the nucleotide sequences of a series of housekeeping gene sequences common to all *Salmonella enterica*

strains to determine the sequence of an individual strain. In the public *Salmonella enterica* database, results can be exchanged easily between laboratories using MLST (mlst. Warwick.ac.UK/mlst/DBS/Senterica/)(Enright and Spratt 1999). Over expression of efflux pump systems, alterations in the cell envelope, down-regulation of porins, and an increased lipopolysaccharide (LPS) component of the outer membrane can all contribute to this pathway (Giraud et al. 2006; Hopkins et al. 2005; Miró et al. 2004). Other processes include iv) quorum sensing (QS) and v) biofilm development (Miró et al. 2004; Giraud et al. 2000; Piddock et al. 1998). Efflux-mediated resistance has been thoroughly researched, and it is often associated with reduction in drug absorption (due to changes in outer membrane permeability) and active drug export (through efflux pumps) in many circumstances (Kumar and Schweizer 2005). These efflux systems are naturally occurred in gram-positive and gram-negative bacteria, as well as eukaryotic organisms, and are involved in the identification and extrusion of hazardous chemicals from cells. However, because of the molecular architecture of the cell membrane, efflux-mediated resistance is a more complicated issue in gram-negative bacteria (Kumar and Schweizer 2005; Nikaido 1998a, b). Thus, the distribution of efflux-mediated drug resistance among *Salmonella* strains is utmost important to understand the mechanisms that occur in a natural selection process for antibiotic resistance (Darby et al. 2022).

Salmonella is most commonly found in hospitals and hospital sewage and poses a significant threat to public health and the healthcare system due to the anticipated rise in migration that could expedite the spread of drug-resistant bacteria. To investigate the serovar and diversity of antimicrobial resistance profiles of isolated *Salmonella* strains, various methods, such as biochemical characterization, PCR, two- and seven-loci MLST, antimicrobial susceptibility testing, and efflux pump activity, were utilized in the present study.

Materials and methods

Sample collection and enrichment

Over a 2-year period (2020–2021), a total of 284 hospital sewage samples were randomly collected in two phases from Bhopal, Madhya Pradesh, India. The samples were enriched in buffered peptone water at 37 °C and 180 rpm for 6 h before being transferred to Rappaport Vassiliadis (RV) at a 1:50 dilution and incubated overnight at 42 °C and 180 rpm. A loopful of overnight RV-culture was streaked on hektoen enteric (HE) agar and plates were incubated at 37 °C overnight. Five colonies showing a black center with a

greenish periphery were chosen for further biochemical and molecular characterization. Gram staining, motility, indole lysine, methyl red, triple sugar iron agar, and urease tests were conducted for the identification of *Salmonella* spp.

Molecular Confirmation of *Salmonella*

The biochemically positive *Salmonella* isolates were further confirmed by employing PCR assay that detects *hilA*, *stn*, *invA*, *typh*, and *iroB* genes specific for the genus, species, and toxin of *Salmonella*. The details of primers are indicated in Table 1. For genomic DNA extraction, *Salmonella* isolates were inoculated in Luria Bertani (LB) broth and incubated at 37 °C, 180 rpm for 16–18 h. The DNA was extracted using a bacterial genomic DNA isolation kit (GeneJET Genomic DNA Purification Kit, Thermo Scientific™). PCR was done using the 25 µl of PCR reaction containing nucleus-free water (10.5 µl), master mixture (12.5 µl), forward and reverse primers (0.5 µl each) and DNA template (1 µl). The PCR amplification was performed with following conditions for *hilA* and *typh* genes were as follows: initial denaturation at 95 °C for 5 min, 34 cycles of 95 °C for 30 s, 53 °C for 90 s, and 72 °C for 90 s, and then a final extension at 72 °C for 10 min. The amplification conditions for *stn* and *invA* genes were as the following: initial denaturation at 95 °C for 5 min, 34 cycles of 95 °C for 30 s, 58 °C for 30 s, and 72 °C for the 30s, and then a final extension at 72 °C for 10 min. The amplification conditions for *iroB*

gene were as the following: initial denaturation at 95 °C for 5 min, 34 cycles of 95 °C for 30 s, 55 °C for 30 s, and 72 °C for 30 s; and then a final extension at 72 °C for 10 min. After completion, the PCR products were resolved on 1% agarose gel stained with ethidium bromide (0.5 µg/ml) and visualized using UV gel documentation system.

Antibiotic resistance profiling

After the biochemical and molecular confirmation, *Salmonella* Typhimurium recovered from HE agar plates were tested for antimicrobial susceptibility by the disc diffusion method described by the Clinical and Laboratory Standards Institute (CLSI) (Clinical and Institute 2017) against the different antimicrobial agents. The antibiotics used in this study were the following: Imipenem (10 µg), Ciprofloxacin (5 µg), Tobramycin (10 µg), Moxifloxacin (5 µg), Ofloxacin (5 µg), Sparfloxacin (5 µg), Levofloxacin (5 µg), Norfloxacin (10 µg), Co-Trimoxazole (25 µg), Colistin (10 µg), Nalidixic acid (30 µg), Augmentin (30 µg), Kanamycin (30 µg), Gatifloxacin (5 µg), Gentamicin (10 µg), Amikacin (30 µg), Streptomycin (10 µg), Ceftriaxone (30 µg), Cefpodoxime (10 µg) and Ticarcillin (75 µg). Briefly, bacterial isolates were emulsified in 1x-PBS and a lawn was prepared on Mueller–Hinton agar plates, then antimicrobial discs were placed on the surface of agar plates with the help of sterile forceps and incubated at 37 °C overnight. *Salmonella* Typhimurium 3232 was used as a control strain.

Table 1 List of primers used in this study

SN	Gene	Primer Sequence (5' to 3')	Annealing Temp (°C)	Amplification size (bp)	Reference
01	<i>hilA</i>	F: CTGCCGAGTGTTAAGGATA R: CTGTGCGCTTAATCGCATGT	53	497	(Guo et al. 2000)
02	<i>stn</i>	F: CTTTGGTCGTAAAATAAGGCG R: TGCCCAAAGCAGAGAGATTC	58	260	(Makino et al. 1999)
03	<i>invA</i>	F:GTGAAATTATCGCCACGTTTCGGGCAA R: TCATCGCACCGTCAAAGGAACC	58	284	(Galán 2001)
04	<i>typh</i>	F: TTGTTCACCTTTTACCCCTGAA R: CCCTGACAGCCGTTAGATATT	53	401	(Alvarez et al. 2004)
05	<i>iroB</i>	F:GCGTATTCTGTTTGTCTGGTCC R:TACGTTCCCACCATCTTCCC	55	606	(Baumler et al. 1997)
06	MLST_aroC	F: CCTGGCACCTCGCGCTATAC R: CCACACACGGATCGTGGCG	55	826	(Kidgell et al. 2002)
07	MLST_dnaN	F:ATGAAATTTACCGTTGAACGTGA R: AATTTCTCATTCGAGAGGATTGC	55	833	(Kidgell et al. 2002)
08	MLST_hemD	F: GAAGCGTTAGTGAGCCGTCTGCG R: ATCAGCGACCTTAATATCTTGCCA	55	666	(Kidgell et al. 2002)
09	MLST_hisD	F: GAAACGTTCCATTCCGCGCAGAC R: CTGAACGGTCATCCGTTTCTG	55	894	(Kidgell et al. 2002)
10	MLST_purE	F: ATGTCTTCCCGCAATAATCC R: TCATAGCGTCCCCCGCGGATC	55	894	(Kidgell et al. 2002)
11	MLST_sucA	F: AGCACCGAAGAGAAACGCTG R: GGTTGTTGATAACGATACGTAC	55	643	(Kidgell et al. 2002)
12	MLST_thrA	F: GTCACGGTGATCGATCCGGT R: CACGATATTGATATTAGCCCG	55	852	(Kidgell et al. 2002)

The antimicrobial susceptibility of *S. Typhimurium* isolates was determined according to the zone diameter breakpoints for *Enterobacteriaceae*. The results of the antibiogram test were recorded as sensitive, intermediate or resistant, and the diameters of the inhibition zones were compared with the diameters of CLSI interpretative standards (Clinical and Institute 2017). *Salmonella* isolates resistant to two or more antimicrobial classes will be considered multidrug-resistant strains (Sweeney et al. 2018).

Multi-locus sequence typing (MLST) analysis

The MLST database can be a powerful tool that unambiguously characterizes bacterial isolates using the sequences of internal fragments of seven housekeeping genes. These genes are essential and known to be evolved more rapidly than 16 S rRNA genes. The complete seven-genes MLST were applied for representative isolates based on the partial sequences of the following seven housekeeping genes: *aroC*, *dnaN*, *hemD*, *hisD*, *purE*, *sucA* and *thrA* by using the gene-specific primers in Table 1 (Kidgell et al. 2002). PCR amplifications were performed in 50 µl of PCR reaction containing nucleus-free water (21 µl), 2x master mixture (25 µl), forward and reverse primer (1 µl each) and 1 µl (roughly 50 ng) DNA as a template. The PCR cycling program is as follows: 10 min 95 °C followed by 30 cycles of 30 s at 95 °C, 30 s at 55 °C and 45 s at 72 °C and completed by a final extension for 10 min at 72 °C. The PCR amplified products were resolved in 1% agarose gels, and amplicons products were purified with a gel extraction kit from Qiagen (K0691 Thermo Scientific™) according to the manufacturer's protocol, and the extracted products were submitted for sequencing at Eurofins genomics India Pvt. Ltd. (India).

The obtained sequences were edited and aligned using BioEdit software and submitted to the *Salmonella* MLST database for allele number assignment and, in the case of the complete seven-gene scheme, for sequence type (ST) assignment (<http://mlst.warwick.ac.uk/mlst/dbs/Senterica>). The data submitted in the MLST database is automatically included in the Enterobase (<http://enterobase.warwick.ac.uk/species/index/senterica>), a much more extensive database containing the data for the MLST database plus STs retrieved from reported complete *Salmonella* genomes. The alleles and STs obtained for Peruvian isolates were compared with those reported in the Enterobase.

Determination of efflux pump activity

Drug efflux is a key resistance mechanism, particularly in gram-negative bacteria. Direct measurement of efflux pump activity can be done by analyzing EtBr accumulation based on fluorescence. Different EtBr concentrations were used in

the cartwheel test. The efflux pump activity of MDR strains of *S. Typhimurium* isolates was determined by the ethidium bromide (EtBr) cartwheel test as described by Martins et al. (2013) with minor modification (Martins et al. 2013). This method allows the comparison of different isolates based on their capacity to pump EtBr out of the cell. The isolates were streaked on solid media Mueller–Hinton agar plates containing increasing concentrations (in the range of 0.2 µg/mL to 4.0 µg/mL) of EtBr. The fluorescence emitted would be inversely proportional to their capacity to extrude the compound and will be compared to the fluorescence of control strain. For this, single isolated colonies of MDR *S. Typhimurium* isolates were inoculated in LB-Broth and incubated at 37 °C, 180 rpm overnight. Afterwards, overnight grown cultures were diluted at 1:100 in fresh LB-broth until they reached an optical density (OD) of 0.5–0.6 at 600 nm. Agar plates containing EtBr concentrations were prepared on the same day of the experiment and protected from direct light. The plates were then divided into cartwheel patterns, and a loopful culture (10⁶ cells/mL) was streaked on EtBr plates. The plates were wrapped in aluminum foil and incubated at 37 °C overnight. After incubation, the plates were examined under the Gel documentation system. The minimum concentration of EtBr (MC_{EtBr}) that produced fluorescence of the bacterial mass was recorded.

Results

Isolation and confirmation of *Salmonella* Typhimurium

Out of 284 samples, 240 (84%) showed positive growth on both the RV broth medium and HE-agar plate. The positive control culture showed green colonies with black centers, as expected for *S. Typhimurium* on HE-agar. Colonies showing positive culture like morphology in replicates (n=5) were selected for biochemical characterization. Out of 240, 160 samples were biochemically confirmed as presumptive *Salmonella* spp. and subjected for further molecular confirmation. Out of 160, 38 colonies were confirmed *S. Typhimurium* isolates through the amplification of the genus-specific *hila*, *stn*, *invA*, *typh* and *iroB* genes with amplicon sizes of 497 bp, 260 bp, 284 bp, 401 bp, and 606 bp, respectively (Fig. 1).

Antimicrobial resistance and antibiogram profiles of the *Salmonella* isolates

The molecularly confirmed *S. Typhimurium* (38) isolates were subjected to the Kirby–Bauer disc diffusion method demonstrating the antibiotic resistance profile and

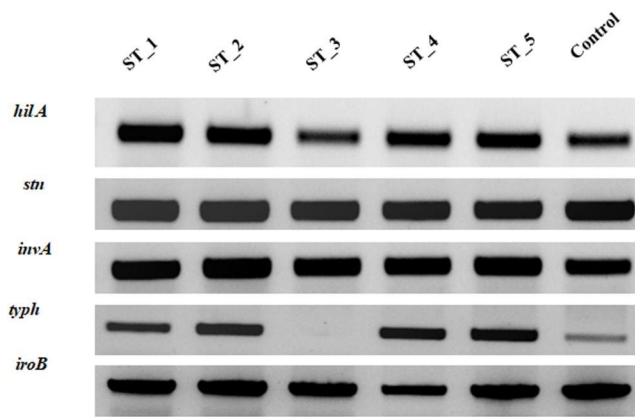


Fig. 1 Molecular confirmation of isolated *Salmonella* strains. The PCR products were ran on 1% agarose gel. Row *hilA*~497 bp, row *stn*~260 bp, row *invA*~284 bp, row *typh*~401 bp and row *iroB*~606 bp

phenotypes. The antibiogram of *S. Typhimurium* isolates was performed using 20 antibiotics which included imipenem, ciprofloxacin, tobramycin, moxifloxacin, ofloxacin, sparfloxacin, levofloxacin, norfloxacin, co-trimoxazole, colistin, nalidixic acid, augmentin, kanamycin, gatifloxacin, gentamicin, amikacin, streptomycin, ceftriaxone, cefpodoxime and ticarcillin (Table 2). Among 38, 16 strains are resistance less than 2 antibiotics; 5 strains (resistance more than 2 antibiotics) were resistant against most of the antibiotics tested. Five strains showed resistance or intermediate resistance against augmentin, amikacin, streptomycin, imipenem, and moxifloxacin, while either sensitive or intermediate for other targeted antibiotics. None of the isolates

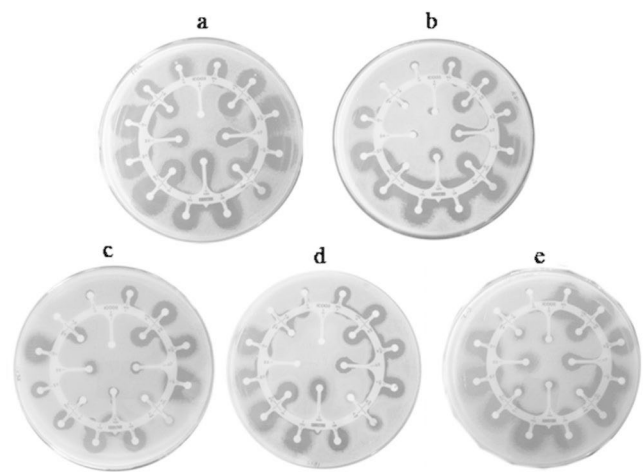


Fig. 2 Disk diffusion method used to determine the antibiogram of an antibiotic resistance (AMR) in the isolated strain of *Salmonella* Typhimurium. Imipenem (IPM) 10 µg, Ciprofloxacin (CIP) 5 µg, Tobramycin (TOB) 10 µg, Moxifloxacin (MO) 5 µg, Ofloxacin (OF) 5 µg, Sparfloxacin (SPX) 5 µg, Levofloxacin (LE) 5 µg, Norfloxacin (NX) 10 µg, Co-Trimoxazole (COT) 25 µg, Colistin (CL) 10 µg, Nalidixic acid (NA) 30 µg, Augmentin (AMC) 30 µg, Kanamycin (K) 30 µg, Gatifloxacin (GAT) 5 µg, Gentamicin (GEN) 10 µg, Amikacin (AK) 30 µg, Streptomycin (S) 10 µg, Ceftriaxone (CTR) 30 µg, Cefpodoxime (CPD) 10 µg and Ticarcillin (TI) 75 µg on the isolated strain of *Salmonella*

have shown resistance towards levofloxacin and gatifloxacin (Fig. 2).

Table 2 Multiple antimicrobial resistance patterns showed by *Salmonella* isolate strains

Antibiotic	Isolated Strains (observed phenotype based on Clinical and Laboratory Standards Institute)					
	ST-1	ST-2	ST-3	ST-4	ST-5	Control
Imipenem (IPM) 10 µg	R	I	R	R	I	S
Ciprofloxacin (CIP) 5 µg	I	I	S	S	S	S
Tobramycin (TOB) 10 µg	S	R	S	I	R	S
Moxifloxacin (MO) 5 µg	R	R	R	R	R	I
Ofloxacin (OF) 5 µg	S	S	S	I	S	S
Sparfloxacin (SPX) 5 µg	I	S	R	R	I	S
Levofloxacin (LE) 5 µg	S	S	S	S	S	S
Norfloxacin (NX) 10 µg	S	S	S	I	S	S
Co-Trimoxazole (COT) 25 µg	R	R	S	R	S	S
Colistin (CL) 10 µg	I	I	S	I	I	S
Nalidixic acid (NA) 30 µg	I	S	S	S	S	S
Augmentin (AMC) 30 µg	R	R	I	R	R	S
Kanamycin (K) 30 µg	R	R	I	R	S	S
Gatifloxacin (GAT) 5 µg	S	S	S	S	S	S
Gentamicin (GEN) 10 µg	S	R	S	R	I	S
Amikacin (AK) 30 µg	R	R	R	R	R	I
Streptomycin (S) 10 µg	R	R	R	R	R	I
Ceftriaxone (CTR) 30 µg	R	I	S	R	I	S
Cefpodoxime (CPD) 10 µg	I	S	S	S	S	S
Ticarcillin (TI) 75 µg	R	R	S	R	R	I

MLST analysis

The nucleotide sequence-based method Multi-Locus Sequence Type (MLST) has been widely used for genotyping bacterial isolates. For *S. enterica*, MLST compares internal sequences of seven housekeeping genes (*aroC*, *dnaN*, *hemD*, *hisD*, *purE*, *sucA* and *thrA*) from 399 to 501 bp. Our goal was to find loci with high variability and discriminatory potential among *Salmonella* serovars on the Entero Base online platform ([HTTP://enterobase.Warwick.ac.UK/species/index/senterica](http://enterobase.warwick.ac.uk/species/index/senterica)). Our analysis found that the *hemD* and *purE* sequences from the seven-locus MLST scheme were the most significant. The resulting sequences were submitted into the *Salmonella* MLST database for allele number assignment. The seven-loci MLST scheme was applied for the selection of the isolates to validate the results obtained by the seven-loci scheme. Out of the 5 MDR isolates, ST138 was found to be the most prevalent STs in the study. All ST138 were found to be *Salmonella enterica* subsp. *enterica* Serovar Typhimurium. Interestingly one ST (ST53) (strain 3) was found to be *S. enterica* serovar Dublin.

Efflux pump activity by EtBr-Agar cartwheel assay

The ability of an organism to remove EtBr from its cells is a key factor contributing to antibiotic resistance. The efflux activities of the isolates were effectively distinguished using EtBr concentrations ranging from 0.2 µg/mL to 4.0 µg/mL. Among the four MDR *S. Typhimurium* isolates, three exhibited pronounced efflux activity as they did not retain EtBr inside their cells (Fig. 3). The capacity of each selected bacterial strain to expel EtBr was inversely proportional to the

fluorescence emitted on Mueller–Hinton agar plates containing EtBr (Fig. 3).

Discussion

In developing countries, particularly India, *Salmonella* is the leading cause of foodborne illness and mortality, with non-typhoidal *Salmonella* (NTS) strains being the most common cause of gastroenteritis globally (Kumawat et al. 2016). The emergence of antibiotic-resistant *Salmonella* spp. has made treatment more challenging. *Salmonella* Typhimurium, a food poisoning pathogen, possesses several virulence factors and is increasingly resistant to antibiotics due to the overuse of antibacterial drugs in clinical settings (Paudyal et al. 2020). According to the World Health Organization (WHO), the most prevalent *Salmonella* serovars responsible for human infections are *Salmonella* Infantis, *Salmonella* Enteritidis, a monophasic variation of *Salmonella* Typhimurium, and *Salmonella* Derby (Ke et al. 2014).

In this study, we successfully isolated four *S. Typhimurium* strains from sewage samples. Antibigram analysis revealed that all four strains were MDR, showing intermediate or resistant characteristics towards augmentin, amikacin, streptomycin, imipenem, and moxifloxacin, while being either sensitive or intermediate towards other targeted antibiotics. Similar studies have been conducted in the past, isolating *Salmonella* strains from various sources such as municipal sewage, hospital sewage, and animal sewage. Studies have shown that *Salmonella* strains isolated from poultry farms can display a broad range of antimicrobial resistance, even if antimicrobials have not been used therapeutically (Liljebjelke et al. 2017). Dutta et al. screened 150 samples and recovered 8% *Salmonella* Typhimurium isolates from fresh pigeons' carcasses. The antibiotic sensitivity of all the isolates was tested, and the results revealed that all isolates showed high resistance towards ampicillin, tetracycline, and nitrofurantoin (Dutta et al. 2013).

Similarly, in recent studies nine *S. Typhimurium* have been isolated from environmental sources of poultry wet markets out of which 88.88% were found to be resistant to 2 or more antibiotic classes. Their results indicated that wet poultry markets environment is the potential contributor of MDR *S. Typhimurium* (Anbazhagan et al. 2019; Salman et al. 2021 isolated 29 as *S. Typhi* and confirmed by using Vitek 2 compact and RT-PCR and isolated strains exhibited resistance to penicillins, cephalosporins, and monobactam (Salman et al. 2021). Another study conducted using caecal and faecal samples, isolated 5 *S. Typhimurium* strains whose antibiotic resistant patterns were examined against 25 antibiotics of different classes, with 100% resistance against oxacillin, penicillin and clindamycin (Mir et al. 2015). It

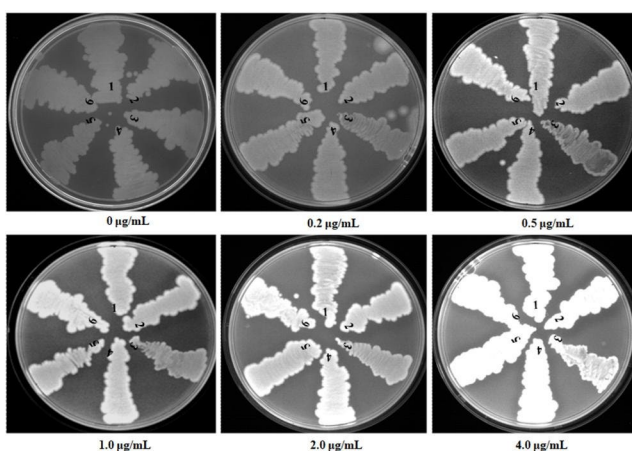


Fig. 3 Ethidium bromide (EtBr) cartwheel method for determining efflux pump activity. An inoculation loopful of fresh culture was inoculated with different EtBr concentrations on agar media. Overnight, the plates were incubated at 37 °C. A UV illuminator was used to visualize and document the plates. Number 1–5 isolated strains and 6 control strains

has been reported that 35% and 10.2% of *S. Typhimurium* strains were isolated from human samples in the years 2016 and 2018, respectively. Elez *et al.* 2021 highest isolation rate (33.1%). *S. Enteritidis* (5.8%), and *S. Typhimurium* (2.8%) isolated from laying hens, table eggs, and humans exhibits resistance to Multidrug-resistant (MDR) detected in 83.8% of the isolates with a multiple antibiotic resistance index of 0.21 to 0.57 (Abou Elez *et al.* 2021). Most of *Salmonella* isolated from duck, pigeon, and quail carcasses marketed in Mansoura, Egypt were showed multidrug-resistant against ampicillin, β -lactamase, cefotaxime and colistin (Elshebrawy *et al.* 2021).

Nair *et al.* have isolated three *S. Typhimurium* from a total of 550 samples, including faecal samples of diarrheagenic infants and adults, fresh vegetables and, sewage which were identified by employing PCR targeting genus-specific *inv A* gene, and Typhimurium-specific *spy* gene (Nair *et al.* 2015). White *et al.* have collected 200 meat samples, out of which two samples were isolated as *Salmonella enterica* serotype Typhimurium that showed resistance against 12 out of 17 antibiotics tested (White *et al.* 2001). M. Kangara *et al.* (2020) targeted *inv A* and *spv C* genes for molecular characterization of *S. Typhimurium* isolated from fecal samples of village chickens. Our findings were comparable to those reported previously in India (Singh *et al.* 2013). Both virulence and antimicrobial resistance genes affect the *S. Typhimurium* pathogenicity (Nikaido *et al.* 2012). Drug resistance can be caused by various factors, including chemical modifications, increased membrane permeability, plasmid-encoded proteins, efflux pumps, and chromosomal modifications. Genetic factors enhance *Salmonella*'s survival by maintaining drug resistance genes that lead to the emergence of MDR strains (Jajere 2019). The molecular characterization of *Salmonella* was done using different genes, which are particular (*hlyA*, *stn*, *invA*, *typh*, and *iroB* genes) for *Salmonella* Typhimurium confirmation. There is no significant variation in the molecular pattern of the four *hlyA*, *stn*, *invA*, and *iroB* genes of identification. However, the *typh* gene showed a different kind of molecular pattern. Several authors have conducted molecular profiling of *Salmonella* in an attempt to *iroB* gene to determine strain types and their distribution. However, the use *iroB* gene as *Salmonella* Typhimurium may not be suitable for the confirmations. Therefore, we suggest *typh* gene is more accurate and precise for molecular confirmation, which our study evidenced.

After molecular confirmations, we performed multi-locus sequence typing (MLST) analysis using seven housekeeping genes (*aroC*, *dnaN*, *hemD*, *hisD*, *purE*, *sucA*, and *thrA*). MLST is a widely accepted molecular typing approach that has been used to characterize several pathogenic bacteria. It has gained recognition as one of the best

approaches available for determining the genetic relatedness and distributions of MDR isolates. We determined the *Salmonella* Typhimurium STs based on a close match of selected MLST alleles. All five MDR isolates were typed using MLST based on their internal sequences of the *Salmonella* seven housekeeping genes. ST138 was the most prevalent ST in our study. It has been observed that MLST has a more remarkable discriminatory ability than gene-specific PCR for typing *Salmonella* confirmations. Strain 5 in our study was confirmed at the molecular level as *Salmonella* Typhimurium (*iroB* gene), but based on MLST, it was found to be only 4. Therefore, MLST could provide an accurate discrimination between different serotypes in *Salmonella*. It has been suggested that *Salmonella* and *E. coli* share a common ancestor that lost its genetic material and was reintroduced as a single fragment from an exogenous source (Kotetishvili *et al.* 2002).

The most common ST observed in our study was ST138, which highlights the remarkable discriminatory ability of MLST compared to gene-specific PCR for *Salmonella* typing. This was exemplified by strain 5, which was identified at the molecular level as *Salmonella* Typhimurium based on the *iroB* gene, but was found to be ST4 based on MLST analysis. Therefore, MLST can accurately differentiate between different *Salmonella* serotypes, as suggested (Achtman *et al.* 2012).

Bacterial efflux systems expel chemically and structurally unrelated compounds in an energy-dependent manner, contributing to the MDR phenotype of bacterial pathogens (Martins *et al.* 2013). Even in EtBr cartwheel test negative isolates, bacterial cells can effectively extrude a maximum concentration of EtBr, which can increase resistance to at least one antibiotic (Anbazhagan *et al.* 2019). Since EtBr is a widely used substrate for efflux pump systems, its use in the cartwheel test method allows for the evaluation of efflux activity using a broad range of efflux pump substrates, making it comparable to other methods for detecting efflux pump activity (Costa *et al.* 2011).

Our study found that three out of four MDR *S. Typhimurium* isolates exhibited pronounced efflux activity, as they did not retain EtBr inside the cells. Similarly, Anbazhagan *et al.* evaluated the role of efflux pumps in AMR using both the ethidium bromide cartwheel test and efflux pump inhibition assay, and found that in MDR *S. Typhimurium* with negative EtBr cartwheel test, efflux pumps may still contribute to resistance for at least one antibiotic. It should be noted that the lack of efflux activity for EtBr may not indicate the total absence of efflux pumps in bacterial strains, as the specificity of efflux pumps for EtBr and antibiotics may differ (Anbazhagan *et al.* 2019). Additionally, *Salmonella enterica* serotype Dublin showed

a different resistance profile and efflux pump activity compared to the other isolated strains.

Drug resistance in bacteria is a complex phenomenon caused by various mechanisms such as decreased outer membrane permeability, changes in lactamase and penicillin-binding proteins, and activation of efflux pumps (Kapoor et al. 2017). However, the limited therapeutic options have created a demand for new strategies to address the issue of antibiotic-resistant bacteria. Inappropriate and extensive use of antibiotics as therapeutic agents and growth promoters in laying hen farms and for treating various human diseases in developing countries has led to the emergence of antimicrobial resistance bacteria at an alarming rate (Abou Elez et al. 2021). The increasing resistance of *Salmonella* to antibiotics demonstrates the diminishing efficacy of antibiotics in treating the disease. The spread of antibiotic-resistant bacteria is further facilitated by mobile genetic elements associated with conjugative plasmids, resulting in the rapid dissemination of antibiotic-resistant genes in the food chain and eventually to humans (Verraes et al. 2013). Therefore, strict measures need to be adopted to control the pathogen's spread, including timely surveillance of pathogenic strains in different environmental compartments and the regulation of proper vaccination protocols for *Salmonella* in areas where resistant strains are prevalent.

Conclusion

This study highlights the prevalence of MDR *Salmonella* Typhimurium in sewage samples and emphasizes the need for preventive measures to prevent its spread. The isolated strains were characterized using biochemical and molecular identification, antimicrobial resistance profiling, MLST, and efflux pump activity. The strains displayed high resistance to several antibiotics, including Amikacin, Streptomycin, Moxifloxacin, Augmentin, and Ticarcillin, indicating a correlation between antimicrobial phenotypes and genotypes. Although the prevalence of this serotype was low, its resistance to commonly used antibiotics is concerning. The presence of MDR *Salmonella* in sewage samples suggests its contribution to the environmental resistome. This study underscores the importance of monitoring MDR *Salmonella* in different environmental compartments, including hospital and municipal sewage in addition to poultry farms, to provide early warning of changes in resistance patterns.

Acknowledgements The authors are thankful to the Director ICMR-NIREH, Bhopal for providing the necessary facilities for the current study.

Author contributions MK, BN, PS, NP DKS, SS and MK designed the experiments and carried out the experimental work. MK, BN, PS, NP, DKS, SS, RRT, SS and MK authors were involved in scientific

discussion and analysis of the data. All authors read and approved the final manuscript.

Funding This work carryout under the Indian council of medical research grants no. 3/1/3PDF (20)/2019-HRD (20th batch).

Declarations

Conflict of interest The authors declare that they have no conflict of interest in this work/manuscript.

Competing financial interests Statement The authors declare no competing financial interests.

All data The data that support the findings of this study are available on request from the corresponding author.

Ethics statement This work not involves any live subjects like human or animal.

References

- Abou Elez RMM, Elsohaby I, El-Gazzar N, Tolba HMN, Abdelfatah EN, Abdellatif SS et al (2021) Antimicrobial Resistance of *Salmonella* enteritidis and *Salmonella* typhimurium isolated from laying hens, table Eggs, and humans with respect to antimicrobial activity of Biosynthesized Silver Nanoparticles. Anim (Basel) 11(12). <https://doi.org/10.3390/ani11123554>
- Achtman M, Wain J, Weill F-X, Nair S, Zhou Z, Sangal V et al (2012) Multilocus sequence typing as a replacement for serotyping in *Salmonella* enterica. PLoS pathogens, 8(6), e1002776
- Alvarez J, Sota M, Vivanco AB, Perales I, Cisterna R, Rementeria A et al (2004) Development of a multiplex PCR technique for detection and epidemiological typing of *Salmonella* in human clinical samples. J Clin Microbiol 42(4):1734–1738. <https://doi.org/10.1128/JCM.42.4.1734-1738.2004>
- Anbazzhagan PV, Thavitiki PR, Varra M, Annamalai L, Putturu R, Lakshini VR et al (2019) Evaluation of efflux pump activity of multidrug-resistant *Salmonella* Typhimurium isolated from poultry wet markets in India. Infect Drug Resist 12:1081–1088. <https://doi.org/10.2147/IDR.S185081>
- Bakshi CS, Singh VP, Malik M, Singh RK, Sharma B (2003) 55 kb plasmid and virulence-associated genes are positively correlated with *Salmonella* enteritidis pathogenicity in mice and chickens. Vet Res Commun 27(6):425–432. <https://doi.org/10.1023/a:1025720306045>
- Baumler AJ, Heffron F, Reissbrodt R (1997) Rapid detection of *Salmonella* enterica with primers specific for iroB. J Clin Microbiol 35(5):1224–1230. <https://doi.org/10.1128/jcm.35.5.1224-1230.1997>
- Bharadwaj A, Rastogi A, Pandey S, Gupta S, Sohal JS (2022) Multi-drug-Resistant Bacteria: Their mechanism of action and prophylaxis. BioMed Research International, 2022
- Brunelle BW, Bearson BL, Allen HK (2016) Prevalence, evolution, and dissemination of antibiotic resistance in *Salmonella*. Food Borne Pathogens and Antibiotic Resistance
- Chaudhari R, Singh K, Kodgire P (2022) Biochemical and molecular mechanisms of Antibiotic Resistance in *Salmonella* spp. Research in Microbiology, 103985
- Clinical, Institute LS (2017) Performance standards for antimicrobial susceptibility testing. Clinical and Laboratory Standards Institute Wayne, PA

- Costa SS, Falcao C, Viveiros M, Machado D, Martins M, Melo-Cristino J et al (2011) Exploring the contribution of efflux on the resistance to fluoroquinolones in clinical isolates of *Staphylococcus aureus*. *BMC Microbiol* 11:241. <https://doi.org/10.1186/1471-2180-11-241>
- Darby EM, Trampari E, Siasat P, Gaya MS, Alav I, Webber MA et al (2022) Molecular mechanisms of antibiotic resistance revisited. *Nature Reviews Microbiology*, 1–16
- Dutta P, Borah M, Sarmah R, Gangil R (2013) Isolation of *Salmonella* Typhimurium from pigeons (*Columba livia*) in Greater Guwahati, its histopathological impact and antibiogram. *Springer*, pp. 147–150 1, p 22
- Elshebrawy HA, Mahros MA, Abd-Elghany SM, Elgazzar MM, Hayashidani H, Sallam KI (2021) Prevalence and molecular characterization of multidrug-resistant and β -lactamase producing *Salmonella* enterica serovars isolated from duck, pigeon, and quail carcasses in Mansoura, Egypt. *LWT*, 149, 111834
- Enright MC, Spratt BG (1999) Multilocus sequence typing. *Trends Microbiol* 7(12):482–487. [https://doi.org/10.1016/S0966-842X\(99\)01609-1](https://doi.org/10.1016/S0966-842X(99)01609-1)
- Fluit AC (2005) Towards more virulent and antibiotic-resistant *Salmonella*? *FEMS Immunol Med Microbiol* 43(1):1–11. <https://doi.org/10.1016/j.femsim.2004.10.007>
- Fluit AC, Schmitz FJ (1999) Class I integrons, gene cassettes, mobility, and epidemiology. *Eur J Clin Microbiol Infect Dis* 18(11):761–770. <https://doi.org/10.1007/s100960050398>
- Galán JE (2001) *Salmonella* interactions with host cells: type III secretion at work. *Annu Rev Cell Dev Biol* 17(1):53–86
- Giraud E, Cloeckaert A, Kerboeuf D, Chaslus-Dancla E (2000) Evidence for active efflux as the primary mechanism of resistance to ciprofloxacin in *Salmonella* enterica serovar Typhimurium. *Antimicrob Agents Chemother* 44(5):1223–1228
- Giraud E, Baucheron S, Cloeckaert A (2006) Resistance to fluoroquinolones in *Salmonella*: emerging mechanisms and resistance prevention strategies. *Microbes Infect* 8(7):1937–1944. <https://doi.org/10.1016/j.micinf.2005.12.025>
- Greig JD, Ravel A (2009) Analysis of foodborne outbreak data reported internationally for source attribution. *Int J Food Microbiol* 130(2):77–87. <https://doi.org/10.1016/j.ijfoodmicro.2008.12.031>
- Guo X, Chen J, Beuchat LR, Brackett RE (2000) PCR detection of *Salmonella* enterica serotype Montevideo in and on raw tomatoes using primers derived from *hlyA*. *Appl Environ Microbiol* 66(12):5248–5252
- Herrera-Leon S, Ramiro R, Arroyo M, Diez R, Usera MA, Echeita MA (2007) Blind comparison of traditional serotyping with three multiplex PCRs for the identification of *Salmonella* serotypes. *Res Microbiol* 158(2):122–127. <https://doi.org/10.1016/j.resmic.2006.09.009>
- Hopkins KL, Davies RH, Threlfall EJ (2005) Mechanisms of quinolone resistance in *Escherichia coli* and *Salmonella*: recent developments. *Int J Antimicrob Agents* 25(5):358–373. <https://doi.org/10.1016/j.ijantimicag.2005.02.006>
- Jajere SM (2019) A review of *Salmonella* enterica with particular focus on the pathogenicity and virulence factors, host specificity and antimicrobial resistance including multidrug resistance. *Veterinary world* 12(4):504
- Kapoor G, Saigal S, Elongavan A (2017) Action and resistance mechanisms of antibiotics: a guide for clinicians. *J Anaesthesiol Clin Pharmacol* 33(3):300–305. https://doi.org/10.4103/joacp.JOACP_349_15
- Ke B, Sun J, He D, Li X, Liang Z, Ke C-w (2014) Serovar distribution, antimicrobial resistance profiles, and PFGE typing of *Salmonella* enterica strains isolated from 2007–2012 in Guangdong, China. *BMC Infect Dis* 14(1):1–10
- Kidgell C, Reichard U, Wain J, Linz B, Torpdahl M, Dougan G et al (2002) *Salmonella* typhi, the causative agent of typhoid fever, is approximately 50,000 years old. *Infect Genet Evol* 2(1):39–45
- Kotetishvili M, Stine OC, Kreger A, Morris JG Jr, Sulakvelidze A (2002) Multilocus sequence typing for characterization of clinical and environmental *Salmonella* strains. *J Clin Microbiol* 40(5):1626–1635. <https://doi.org/10.1128/JCM.40.5.1626-1635.2002>
- Kumar A, Schweizer HP (2005) Bacterial resistance to antibiotics: active efflux and reduced uptake. *Adv Drug Deliv Rev* 57(10):1486–1513. <https://doi.org/10.1016/j.addr.2005.04.004>
- Kumar M, Sarma DK, Shubham S, Kumawat M, Verma V, Nina PB et al (2021) Futuristic non-antibiotic therapies to Combat Antibiotic Resistance: a review. *Front Microbiol* 12:609459. <https://doi.org/10.3389/fmicb.2021.609459>
- Kumawat M, Pesingi PK, Agarwal RK, Goswami TK, Mahawar M (2016) Contribution of protein isoaspartate methyl transferase (PIMT) in the survival of *Salmonella* Typhimurium under oxidative stress and virulence. *Int J Med Microbiol* 306(4):222–230. <https://doi.org/10.1016/j.ijmm.2016.04.005>
- Kumawat M, Chaudhary D, Nabi B, Kumar M, Sarma DK, Shubham S et al (2021) Purification and characterization of Cyclophilin: a protein associated with protein folding in *Salmonella* Typhimurium. *Arch Microbiol* 203(9):5509–5517. <https://doi.org/10.1007/s00203-021-02519-9>
- Leverstein-van Hall MA, Blok M, Donders HET, Paauw AR, Fluit A, A. C., Verhoef J (2003) Multidrug resistance among Enterobacteriaceae is strongly associated with the presence of integrons and is independent of species or isolate origin. *J Infect Dis* 187(2):251–259
- Liljebjelke KA, Hofacre CL, White DG, Ayers S, Lee MD, Maurer JJ (2017) Diversity of Antimicrobial Resistance Phenotypes in *Salmonella* isolated from Commercial Poultry Farms. *Front Vet Sci* 4:96. <https://doi.org/10.3389/fvets.2017.00096>
- Majowicz S, Musto J, Scallan E, Angulo F, Kirk M, O'brien S et al (2010) International Collaboration on Enteric Disease “Burden of Illness” Studies. 2010. *The global burden of nontyphoidal Salmonella gastroenteritis*. *Clin Infect Dis*, 50(6), 882–889
- Makino S, Kurazono H, Chongsangum M, Hayashi H, Cheun H, Suzuki S et al (1999) Establishment of the PCR system specific to *Salmonella* spp. and its application for the inspection of food and fecal samples. *J Vet Med Sci* 61(11):1245–1247. <https://doi.org/10.1292/jvms.61.1245>
- Marshall BM, Levy SB (2011) Food animals and antimicrobials: impacts on human health. *Clin Microbiol Rev* 24(4):718–733. <https://doi.org/10.1128/CMR.00002-11>
- Martin LJ, Fyfe M, Doré K, Buxton JA, Pollari F, Henry B et al (2004) Increased burden of illness associated with antimicrobial-resistant *Salmonella* enterica serotype typhimurium infections. *J Infect Dis* 189(3):377–384
- Martins M, McCusker MP, Viveiros M, Couto I, Fanning S, Pages JM et al (2013) A simple method for Assessment of MDR Bacteria for over-expressed efflux pumps. *Open Microbiol J* 7:72–82. <https://doi.org/10.2174/1874285801307010072>
- Mezal EH, Sabol A, Khan MA, Ali N, Stefanova R, Khan AA (2014) Isolation and molecular characterization of *Salmonella* enterica serovar enteritidis from poultry house and clinical samples during 2010. *Food Microbiol* 38:67–74. <https://doi.org/10.1016/j.fm.2013.08.003>
- Mir IA, Kashyap SK, Maherchandani S (2015) Isolation, serotype diversity and antibiogram of *Salmonella* enterica isolated from different species of poultry in India. *Asian Pac J Trop Biomed* 5(7):561–567
- Miró E, Vergés C, García I, Mirelis B, Navarro F, Coll P et al (2004) Resistance to quinolones and beta-lactams in *Salmonella* enterica due to mutations in topoisomerase-encoding genes, altered cell

- permeability and expression of an active efflux system. *Enferm Infecc Microbiol Clin* 22(4):204–211
- Nair A, Balasaravanan T, Malik SV, Mohan V, Kumar M, Vergis J et al (2015) Isolation and identification of *Salmonella* from diarrheagenic infants and young animals, sewage waste and fresh vegetables. *Vet World* 8(5):669–673. <https://doi.org/10.14202/vetworld.2015.669-673>
- Nikaido H (1998a) Antibiotic resistance caused by gram-negative multidrug efflux pumps. *Clin Infect Dis* 27(Suppl 1):S32–41. <https://doi.org/10.1086/514920>
- Nikaido H (1998b) Multiple antibiotic resistance and efflux. *Curr Opin Microbiol* 1(5):516–523. [https://doi.org/10.1016/s1369-5274\(98\)80083-0](https://doi.org/10.1016/s1369-5274(98)80083-0)
- Nikaido E, Giraud E, Baucheron S, Yamasaki S, Wiedemann A, Okamoto K et al (2012) Effects of indole on drug resistance and virulence of *Salmonella* enterica serovar Typhimurium revealed by genome-wide analyses. *Gut pathogens* 4(1):1–13
- Paudyal N, Pan H, Wu B, Zhou X, Zhou X, Chai W et al (2020) Persistent Asymptomatic Human Infections by *Salmonella* enterica Serovar Newport in China. *mSphere* 5(3). <https://doi.org/10.1128/mSphere.00163-20>
- Piddock LJ, Ricci V, McLaren I, Griggs DJ (1998) Role of mutation in the *gyrA* and *parC* genes of nalidixic-acid-resistant *Salmonella* serotypes isolated from animals in the United Kingdom. *J Antimicrob Chemother* 41(6):635–641. <https://doi.org/10.1093/jac/41.6.635>
- Salman HA, Abdulmohsen AM, Fali MN, Romi ZM (2021) Detection of multidrug-resistant *Salmonella* enterica subsp. enterica serovar typhi isolated from iraqi subjects. *Vet World* 14(7):1922–1928. <https://doi.org/10.14202/vetworld.2021.1922-1928>
- Shah DH, Paul NC, Sischo WC, Crespo R, Guard J (2017) Population dynamics and antimicrobial resistance of the most prevalent poultry-associated *Salmonella* serotypes. *Poult Sci* 96(3):687–702
- Singh R, Yadav A, Tripathi V, Singh R (2013) Antimicrobial resistance profile of *Salmonella* present in poultry and poultry environment in north India. *Food Control* 33(2):545–548
- Sweeney MT, Lubbers BV, Schwarz S, Watts JL (2018) Applying definitions for multidrug resistance, extensive drug resistance and pandrug resistance to clinically significant livestock and companion animal bacterial pathogens. *J Antimicrob Chemother* 73(6):1460–1463
- Varma JK, Greene KD, Ovitt J, Barrett TJ, Medalla F, Angulo FJ (2005) Hospitalization and antimicrobial resistance in *Salmonella* outbreaks, 1984–2002. *Emerg Infect Dis* 11(6):943–946. <https://doi.org/10.3201/eid1106.041231>
- Verraes C, Van Boxtael S, Van Meervenne E, Van Coillie E, Butaye P, Catry B et al (2013) Antimicrobial resistance in the food chain: a review. *Int J Environ Res Public Health* 10(7):2643–2669. <https://doi.org/10.3390/ijerph10072643>
- Vidic J, Manzano M, Raj VS, Pandey RP, Chang C-M (2022) Comparative meta-analysis of antimicrobial resistance from different food sources along with one health approach in Italy and Thailand. *One Health*, 100477
- Wang Z, Zhang Y, Xu H, Chu C, Wang J, Jiao X et al (2022) Whole-genome sequencing analysis reveals pig as the main reservoir for persistent evolution of *Salmonella* enterica serovar Rissen causing human salmonellosis. *Food Res Int* 154:111007
- White DG, Zhao S, Sudler R, Ayers S, Friedman S, Chen S et al (2001) The isolation of antibiotic-resistant *Salmonella* from retail ground meats. *N Engl J Med* 345(16):1147–1154. <https://doi.org/10.1056/NEJMoa010315>
- Wiesner M, Zaidi MB, Calva E, Fernandez-Mora M, Calva JJ, Silva C (2009) Association of virulence plasmid and antibiotic resistance determinants with chromosomal multilocus genotypes in mexican *Salmonella* enterica serovar typhimurium strains. *BMC Microbiol* 9:131. <https://doi.org/10.1186/1471-2180-9-131>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.